

## 7<sup>th</sup> Grade Unit 3 Part 1 Assessment Guide

### General Information

Units 1 and 2 work together to strengthen students' understanding of rational numbers. Part 1 of this assessment focuses on procedural fluency of operations found in MA.7.NSO.2.2. This assessment does not have to be given after lesson 4 of unit 3. It could be given at the end of unit 3 or at some other time when the teacher feels students have had sufficient time to grow in their fluency.

The expectations of procedural fluency for grade 7 students builds on the fluency expectations of grade 6.

"In grade 6, students performed operations with integers, multiplied and divided positive multi-digit numbers with decimals to the thousandths and computed products and quotients of positive fractions by positive fractions, including mixed numbers with procedural fluency."

*-from the Grade 7 B.E.S.T. Instructional Guide for Mathematics*

Because this is the expectation of grade 6 students, the procedural fluency expectations by the end of grade 7 build on this so that students in grade 7 should be able to perform all four operations with rational numbers, including negative values, with procedural fluency. The draft test item specifications include the assessment limit: "Values in expressions must incorporate at least one negative value and/or rational numbers represented in various forms." This assessment limit ensures that what is assessed is not at the sixth-grade fluency expectation. Form A questions 1 and 5, form B questions 2, 9 and 10, and form C questions 2 and 6 use rational numbers in different forms. The other questions incorporate at least one negative value.

This assessment guide provides ideas on ways to continue to build student fluency both by using questions that are similar to the questions on the assessment and how to use the assessment to help a student build to fluency. To be fluent in another language means "having the ability to speak or write that language easily and accurately."

If you are a speaker who is fluent in another language, then you "hear something being said in that other language and you automatically understand what it means." You are not translating words in your mind as you speak but simply know the language and how to use it. This same idea can be applied to fluency in mathematics. Mathematical fluency is being able to manipulate numbers easily and accurately.

Procedural fluency of rational numbers is one of the five areas of emphasis of learning in grade 7. Students will reach this fluency at different times and at different rates. Teachers are encouraged to continue to support student fluency throughout the year by utilizing instructional routines that help students grow in their flexibility of recognizing equivalent forms of rational numbers and performing operations with rational numbers. Utilizing the calculator as a tool, not as a crutch, will require a delicate balance throughout grade 7.

The National Council of Teachers of Mathematics (NCTM) has the following position on procedural fluency.

*Procedural fluency is a critical component of mathematical proficiency. Procedural fluency is the ability to apply procedures accurately, efficiently, and flexibly; to transfer procedures to different problems and contexts; to build or modify procedures from other procedures; and to recognize when one strategy or procedure is more appropriate to apply than another. To develop procedural fluency, students need experience in integrating concepts and procedures and building on familiar procedures as they create their own informal strategies and procedures. Students need opportunities to justify both informal strategies and commonly used procedures mathematically, to support and justify their choices of appropriate procedures, and to strengthen their understanding and skill through distributed practice.*

Please note that NCTM did not state that fluency involves speed, but it did state that it involves efficiency. Efficiency can look different for each individual student. Student confidence can be built by having students share different strategies by using instructional routines such as “or you could,” where students share their different strategies or equivalent results. Another instructional routine that can be used is “why not,” where students examine a strategy that is not the most efficient and explain why the strategy is not a good fit. These and other instructional routines help students build their mathematical agency and flexibility.

Teachers are encouraged to use this assessment to determine which students have reached procedural fluency on all four operations with rational numbers and which students need further support on this procedural fluency. It is suggested that teachers look at student work to determine whether fluency is being hampered by misconceptions or incorrect methods. This assessment guide includes different possible methods of determining the value of the given expression. Students may use other methods that are not shown. The guide purposefully shows these methods so the teacher can consider alternatives. The

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|                                 | <p>teacher may want to consider providing students with “student papers” with questions that are similar to the ones on one of the forms of the Unit 3, Part 1 assessment. The “student papers” could show work using similar methods shown in this guide. Some of the papers could have a minor error. Students can then comment on the accuracy and the efficiency of the student work. This may be helpful to do before having students take this assessment.</p> <p>Multiple forms are provided should teachers wish to have students take different forms or for potential re-assessment at different points in the year once students have had time to grow in their fluency.</p>                        |
| <b>Calculator Information</b>   | The questions on this assessment are no calculator.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Total Recommended Points</b> | The total recommended points in this guide is 100 points. Teachers are encouraged to consider that students who show fluency and understanding despite making minor mistakes receive the full 100 points. If students struggle on the assessment, then teachers are encouraged to allow students to show their growth after remediation by taking other forms of this assessment. Teachers can determine what fluency looks like in their classroom and grade this assessment accordingly. The 10 points for each question is based on 100 points. Teachers may also utilize 5 points per question and set a score such as 85 and above as showing fluency. These are suggestions for the teacher to consider. |

[Form A](#)

[Form B](#)

[Form C](#)

# Form A

| Question | 1 | Benchmark(s)                                               | MA.7.NSO.2.2                                                                                                                                                                                                                                                                                                                                                  |
|----------|---|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |   | <b>Recommended Scoring</b><br><br><b>(10 Total Points)</b> | Consider following student work to determine if a mistake is a minor miscalculation, a major error, or a misconception. All equivalent answers should be expected. The possible methods shown are likely the most common methods that students use. But students who choose other methods and correctly determine an equivalent value should not lose points. |
|          |   |                                                            | <b>Additional Information</b><br><br>This question is assessing student flexibility with different forms of rational numbers. Teachers should look for student understanding of equivalency.                                                                                                                                                                  |
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| Question                                                                                                                                                                                                                                            | 2                                                                                                                                                                                         | Benchmark(s)                                                      | MA.7.NSO.2.2                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <p><b>Using <math>\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}</math></b></p> $-4\frac{5}{24} - \frac{21}{16}$ $\frac{-101}{24} + \frac{-21}{16}$ $\frac{-101 \cdot 16 + -21 \cdot 24}{24 \cdot 16}$ $\frac{-1616 + -504}{384}$ $\frac{-2120}{384}$ | <p><b>Using a common denominator of 48</b></p>                                                                                                                                            | <p><b>Recommended Scoring</b></p> <p><b>(10 Total Points)</b></p> | <p>Consider following student work to determine if a mistake is a minor miscalculation, a major error, or a misconception. All equivalent answers should be expected. The possible methods shown are likely the most common methods that students use. But students who choose other methods and correctly determine an equivalent value should not lose points.</p>                                            |
|                                                                                                                                                                                                                                                     | $-4\frac{5}{24} - \frac{21}{16}$ $-4\frac{5}{24} = -4\frac{10}{48}$ $\frac{21}{16} = 1\frac{5}{16}$ $1\frac{5}{16} = 1\frac{15}{48}$ $-4\frac{10}{48} - 1\frac{15}{48}$ $-5\frac{25}{48}$ | <p><b>Additional Information</b></p>                              | <p>Students who use the method, <math>\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}</math>, may not “reduce” to lowest terms. The answer as shown is sufficient. Simplifying fractions to lowest terms is not learned until grade 6, MA.6.NSO.3.1, and is taught as finding common factors of the numerator and denominator. Students who use a common denominator may not use 48 as the common denominator.</p> |

| Question                                                                                                                                                                                                                                           | 3 | Benchmark(s)                                                | MA.7.NSO.2.2                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div> <div>Using Decimals</div> <math display="block">-4.63 \cdot 3.9</math> <math display="block">\begin{array}{r} 4.63 \\ \times 3.9 \\ \hline 4167 \\ 13890 \\ \hline 18.057 \\ -18.057 \end{array}</math> </div>                               |   | <div>Recommended Scoring</div> <div>(10 Total Points)</div> | <p>Consider following student work to determine if a mistake is a minor miscalculation, a major error, or a misconception. All equivalent answers should be expected. The possible methods shown are likely the most common methods that students use. But students who choose other methods and correctly determine an equivalent value should not lose points.</p> |
| <div> <div>Using Fractions</div> <math display="block">-4\frac{63}{100} \cdot 3\frac{9}{10}</math> <math display="block">\frac{-463}{100} \cdot \frac{39}{10} = \frac{-18057}{1000}</math> <math display="block">\frac{-18057}{1000}</math> </div> |   | <div>Additional Information</div>                           | <p>Two possible methods are shown.</p>                                                                                                                                                                                                                                                                                                                               |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 4 | Benchmark(s)                                                | MA.7.NSO.2.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div> <div>Using Long Division</div> <math display="block">32.5892 \div (-7.91)</math> <math display="block">7.91 \overline{) 32.5892}</math> <math display="block">791 \overline{) 3258.92}</math> <math display="block">\begin{array}{r} 4.12 \\ 791 \overline{) 3258.92} \\ - 3164 \phantom{00} \\ \hline 949 \phantom{00} \\ - 791 \phantom{00} \\ \hline 1582 \phantom{00} \\ - 1582 \phantom{00} \\ \hline 0 \end{array}</math> <p><b>-4.12</b></p> </div>                                                                              |   | <div>Recommended Scoring</div> <div>(10 Total Points)</div> | <p>Consider following student work to determine if a mistake is a minor miscalculation, a major error, or a misconception.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <div> <div>Using Fractions</div> <math display="block">32.5892 \div (-7.91)</math> <math display="block">32 \frac{5892}{10000} \div -7 \frac{91}{100}</math> <math display="block">\frac{325892}{10000} \div \frac{-791}{100}</math> <math display="block">\frac{325892}{10000} \cdot \frac{100}{-791}</math> <math display="block">\frac{32589200}{-7910000}</math> <p><b><math>-\frac{32589200}{7910000}</math> or <math>-\frac{325892}{79100}</math> or <math>-\frac{412}{100}</math></b></p> <p><b>or any other equivalent</b></p> </div> |   | <div>Additional Information</div>                           | <p>The two methods shown are representative of how flexible the benchmark allows students to be. Teachers often ask about the expectations of statewide assessments. The draft test item specifications do not require that answers be in the most simplified form so students should be able to answer equation editor questions without simplification. Multiple choice questions may require students to recognize equivalent responses which means students are applying their knowledge from MA.7.NSO.1.2. Since this is requiring knowledge of two different benchmarks it is not likely students would have to simplify when a question is assessing their knowledge of MA.7.NSO.2.2 because Florida's assessments assess one benchmark for each question.</p> |

| Question                                                                                                                                                                  |  | 5                                                                                                                                                                                                                                    | Benchmark(s)                                                | MA.7.NSO.2.2                                                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>Using Fractions</div> $2.75 \cdot \frac{19}{8}$ $2\frac{3}{4} \cdot \frac{19}{8}$ $\frac{11}{4} \cdot \frac{19}{8}$ $\frac{209}{32} \text{ or any other equivalent}$ |  | <div>Using Decimals</div> $2.75 \cdot \frac{19}{8}$ $\frac{19}{8} = 2\frac{3}{8}$ $\frac{3}{8} = 0.375$ $2.75 \cdot 2.375$ $\begin{array}{r} 2.375 \\ \times 2.75 \\ \hline 11875 \\ 166250 \\ 475000 \\ \hline 6.53125 \end{array}$ | <div>Recommended Scoring</div> <div>(10 Total Points)</div> | Consider following student work to determine if a mistake is a minor miscalculation, a major error, or a misconception.                                 |
|                                                                                                                                                                           |  |                                                                                                                                                                                                                                      | <div>Additional Information</div>                           | This question is assessing student flexibility with different forms of rational numbers. Teachers should look for student understanding of equivalency. |



| Question                                                                                                                                                                                                                                                                                                                                                                                       | 6                                                                                                                                                                                                                                                                                                                                                                                          | Benchmark(s)                                                                                                                                       | MA.7.NSO.2.2                                                                                                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div data-bbox="226 207 558 250">Using Fractions</div> $8\frac{2}{3} \div \left(-\frac{104}{33}\right)$ $\frac{26}{3} \div \left(-\frac{104}{33}\right)$ $\frac{26}{3} \cdot \left(-\frac{33}{104}\right)$ $\frac{\overset{1}{\cancel{26}}}{\underset{1}{\cancel{3}}} \cdot -\frac{\overset{11}{\cancel{33}}}{\underset{4}{\cancel{104}}}$ <p><math>-\frac{11}{4}</math> or any equivalent</p> | <div data-bbox="873 207 1197 250">Using Decimals</div> $8\frac{2}{3} \div \left(-\frac{104}{33}\right)$ $8.\overline{6} \div -3.\overline{15}$ $3.\overline{15} \overline{)8.66666}$ $\begin{array}{r} 2.75 \\ 315 \overline{)866.666} \\ \underline{-630} \phantom{00} \\ 2366 \phantom{00} \\ \underline{-2205} \phantom{00} \\ 1616666\ldots \\ \underline{-1616666\ldots} \end{array}$ | <div data-bbox="1398 207 1665 396">Recommended Scoring<br/>(10 Total Points)</div> <div data-bbox="1419 841 1644 909">Additional Information</div> | <p>Consider following student work to determine if a mistake is a minor miscalculation, a major error, or a misconception.</p> <p>The key shows students using common factors to “simplify” the answer. Students are not required to do this.</p> |

| Question                                                                                                                                                       |  | 7                                                                                                 | Benchmark(s)                             | MA.7.NSO.2.2                                                                                                                                                                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---------------------------------------------------------------------------------------------------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $0.0558$                                                                                                                                                       |  |                                                                                                   | Recommended Scoring<br>(10 Total Points) | Consider following student work to determine if a mistake is a minor miscalculation, a major error, or a misconception.                                                      |
|                                                                                                                                                                |  |                                                                                                   | Additional Information                   | Watch for student misconceptions based on place value. Students are not likely to determine the sum using fractions, but they should have the freedom to do so if they wish. |
| Question                                                                                                                                                       |  | 8                                                                                                 | Benchmark(s)                             | MA.7.NSO.2.2                                                                                                                                                                 |
| Using Fractions                                                                                                                                                |  | Using Decimals                                                                                    | Recommended Scoring<br>(10 Total Points) | Consider following student work to determine if a mistake is a minor miscalculation, a major error, or a misconception.                                                      |
| $-3\frac{2}{3} - \left(-\frac{5}{6}\right)$<br><br>$-\frac{11}{3} + \frac{5}{6}$<br><br>$-\frac{22}{6} + \frac{5}{6}$<br><br>$-\frac{17}{6}$ or any equivalent |  | $-3\frac{2}{3} - \left(-\frac{5}{6}\right)$<br><br>$-3.\bar{6} + 0.8\bar{3}$<br><br>$-2.8\bar{3}$ |                                          |                                                                                                                                                                              |
|                                                                                                                                                                |  |                                                                                                   | Additional Information                   | As part of fluency, students may recall the decimal equivalent of a fraction. For these students, using decimals may be the most efficient method.                           |

| Question                                                                                                                                   | 9                                                                                                                                                              | Benchmark(s)                                               | MA.7.NSO.2.2                                                                                                                                                                                                                                            |
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|                                                                                                                                            |                                                                                                                                                                | <b>Recommended Scoring</b><br><br><b>(10 Total Points)</b> | Consider following student work to determine if a mistake is a minor miscalculation, a major error, or a misconception.                                                                                                                                 |
| <b>Using a common denominator of 32</b>                                                                                                    | <b>Using <math>\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}</math></b>                                                                                         |                                                            | <b>Additional Information</b><br><br>Students may choose to use a common denominator other than 32. Teachers are cautioned against forming an opinion on efficiency. Efficiency can be a personal choice based on each person’s skills and preferences. |
| $-6\frac{17}{32} + \frac{35}{8}$<br><br>$-6\frac{17}{32} + 4\frac{3}{8}$<br><br>$-6\frac{17}{32} + 4\frac{12}{32}$<br><br>$-2\frac{5}{32}$ | $-6\frac{17}{32} + \frac{35}{8}$<br><br>$\frac{-209}{32} + \frac{35}{8}$<br><br>$\frac{-1672 + 1120}{256}$<br><br>$\frac{-552}{256} \text{ or any equivalent}$ |                                                            |                                                                                                                                                                                                                                                         |

| Question                            | 10                                   | Benchmark(s)                                 | MA.7.NSO.2.2                                                                                                                                                                                                              |
|-------------------------------------|--------------------------------------|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     |                                      | Recommended Scoring<br><br>(10 Total Points) | Consider following student work to determine if a mistake is a minor miscalculation, a major error, or a misconception.                                                                                                   |
|                                     |                                      |                                              |                                                                                                                                                                                                                           |
| Fractions                           | Decimals                             | Additional Information                       | This question is an excellent example where students with flexible thinking may find using decimals are more efficient than fractions. Using a similar question for student discussion may help to build student fluency. |
| $-\frac{34}{5} \cdot 1\frac{1}{10}$ | $-\frac{34}{5} \cdot 1\frac{1}{10}$  |                                              |                                                                                                                                                                                                                           |
| $-\frac{34}{5} \cdot \frac{11}{10}$ | $-6\frac{4}{5} \cdot 1\frac{1}{10}$  |                                              |                                                                                                                                                                                                                           |
| $-\frac{374}{50}$                   | $-6\frac{8}{10} \cdot 1\frac{1}{10}$ |                                              |                                                                                                                                                                                                                           |
|                                     | $-6.8 \cdot 1.1$                     |                                              |                                                                                                                                                                                                                           |
|                                     | $-7.48$                              |                                              |                                                                                                                                                                                                                           |

## Form B

| Question                                                                                                                      | 1                                                                                                                                                               | Benchmark(s)                                 | MA.7.NSO.2.2                                                                                                                                                                                                                                                                                                                                                  |
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|                                                                                                                               |                                                                                                                                                                 | Recommended Scoring<br><br>(10 Total Points) | Consider following student work to determine if a mistake is a minor miscalculation, a major error, or a misconception. All equivalent answers should be expected. The possible methods shown are likely the most common methods that students use. But students who choose other methods and correctly determine an equivalent value should not lose points. |
| Using Decimals                                                                                                                | Using Fractions                                                                                                                                                 |                                              |                                                                                                                                                                                                                                                                                                                                                               |
| $-0.036 \cdot 2.9$<br><br>$\begin{array}{r} 0.036 \\ \times 2.9 \\ \hline 324 \\ +720 \\ \hline .1044 \\ -0.1044 \end{array}$ | $-0.036 \cdot 2.9$<br><br>$-\frac{36}{1000} \cdot 2\frac{9}{10}$<br><br>$\frac{-36}{1000} \cdot \frac{29}{10} = \frac{-1044}{10000}$<br><br>$\frac{-261}{2500}$ | Additional Information                       | Two possible methods are shown.                                                                                                                                                                                                                                                                                                                               |

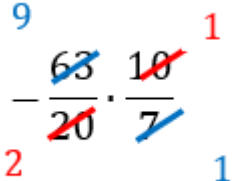
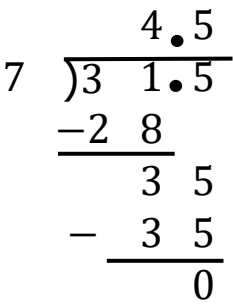
| Question                                                                                                                                                                                                                                                                    | 2                                                                                                                                        | Benchmark(s)                                                                       | MA.7.NSO.2.2                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div data-bbox="226 297 558 337">Using Fractions</div> $4\frac{3}{5} - 0.863$ $4\frac{3}{5} = 4\frac{600}{1000}$ $0.863 = \frac{863}{1000}$ $4\frac{600}{1000} - \frac{863}{1000}$ $\frac{4600}{1000} - \frac{863}{1000}$ $\frac{3737}{1000} \text{ or } 3\frac{737}{1000}$ | <div data-bbox="873 297 1197 337">Using Decimals</div> $4\frac{3}{5} - 0.863$ $4\frac{3}{5} = 4\frac{6}{10} = 4.6$ $4.6 - 0.863$ $3.737$ | <div data-bbox="1398 475 1665 665">Recommended Scoring<br/>(10 Total Points)</div> | <p>Consider following student work to determine if a mistake is a minor miscalculation, a major error, or a misconception. All equivalent answers should be expected. The possible methods shown are likely the most common methods that students use. But students who choose other methods and correctly determine an equivalent value should not lose points.</p> |
|                                                                                                                                                                                                                                                                             |                                                                                                                                          | <div data-bbox="1421 1117 1640 1187">Additional Information</div>                  | <p>This question is assessing student flexibility with different forms of rational numbers. Teachers should look for student understanding of equivalency.</p>                                                                                                                                                                                                       |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Benchmark(s)                                                | MA.7.NSO.2.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| <div data-bbox="73 248 716 1328"> <div>Using Long Division</div> <math display="block">-38.4302 \div 1.27</math> <math display="block">1.27 \overline{)38.4302}</math> <math display="block">127 \overline{)3843.02}</math> <math display="block">127 \overline{)3843.02}</math> <math display="block">\begin{array}{r} 30.26 \\ 127 \overline{)3843.02} \\ \underline{-381} \phantom{00} \\ 3302 \\ \underline{-254} \phantom{00} \\ 762 \\ \underline{-762} \\ 0 \end{array}</math> <p><b>-30.26</b></p> </div> | <div data-bbox="716 248 1358 1328"> <div>Using Fractions</div> <math display="block">-38.4302 \div 1.27</math> <math display="block">-38 \frac{4302}{10000} \div 1 \frac{27}{100}</math> <math display="block">\frac{-384302}{10000} \div \frac{127}{100}</math> <math display="block">\frac{-384302}{10000} \cdot \frac{100}{127}</math> <math display="block">\frac{-38430200}{1270000}</math> <p><math>-\frac{38430200}{1270000}</math> or <math>-\frac{384302}{12700}</math> or <math>-\frac{3026}{100}</math></p> <p><b>or any other equivalent</b></p> </div> | <div>Recommended Scoring</div> <div>(10 Total Points)</div> | <p>Consider following student work to determine if a mistake is a minor miscalculation, a major error, or a misconception.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <div>Additional Information</div>                           | <p>The two methods shown are representative of how flexible the benchmark allows students to be. Teachers often ask about the expectations of statewide assessments. The draft test item specifications do not require that answers be in the most simplified form so students should be able to answer equation editor questions without simplification. Multiple choice questions may require students to recognize equivalent responses which means students are applying their knowledge from MA.7.NSO.1.2. Since this is requiring knowledge of two different benchmarks it is not likely students would have to simplify when a question is assessing their knowledge of MA.7.NSO.2.2 because Florida's assessments assess one benchmark for each question.</p> |

| Question                                                                                                                                      | 4                                                                                                                                                                                                                                                                                                          | Benchmark(s)                                 | MA.7.NSO.2.2                                                                                                                                                                                                              |
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| Using a common denominator of 24                                                                                                              | Using the commutative property                                                                                                                                                                                                                                                                             | Recommended Scoring<br><br>(10 Total Points) | Consider following student work to determine if a mistake is a minor miscalculation, a major error, or a misconception.                                                                                                   |
| $-5\frac{5}{6} + 3\frac{7}{8}$ $-5\frac{20}{24} + 3\frac{21}{24}$ $-\frac{140}{24} + \frac{93}{24}$ $-\frac{47}{24}$ <p>or any equivalent</p> | $-5\frac{5}{6} + 3\frac{7}{8}$ $-5 - \frac{5}{6} + 3 + \frac{7}{8}$ $-5 + 3 - \frac{5}{6} + \frac{7}{8}$ $-5 + 3 - \frac{20}{24} + \frac{21}{24}$ $-2 + \frac{1}{24}$ $-1\frac{24}{24} + \frac{1}{24}$ $-1 - \frac{24}{24} + \frac{1}{24}$ $-1 - \frac{23}{24}$ $-1\frac{23}{24}$ <p>or any equivalent</p> | <p><b>Additional Information</b></p>         | <p>Students may choose to use a common denominator other than 24. Teachers are cautioned against forming an opinion on efficiency. Efficiency can be a personal choice based on each person's skills and preferences.</p> |



| Question                                                                                                                                               |  | 5                                                                                |  | Benchmark(s)                             | MA.7.NSO.2.2                                                                                                                                                                 |
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| Using Fractions                                                                                                                                        |  | Using Decimals                                                                   |  | Recommended Scoring<br>(10 Total Points) | Consider following student work to determine if a mistake is a minor miscalculation, a major error, or a misconception.                                                      |
| $-4\frac{5}{8} - \left(\frac{5}{6}\right)$ $-\frac{37}{8} - \frac{5}{6}$ $-\frac{111}{24} - \frac{20}{24}$ $-\frac{131}{24} \text{ or any equivalent}$ |  | $-4\frac{5}{8} - \left(\frac{5}{6}\right)$ $-4.625 - 0.8\bar{3}$ $-5.458\bar{3}$ |  | Additional Information                   | As part of fluency, students may recall the decimal equivalent of a fraction. For these students using decimals may be the most efficient method.                            |
| Question                                                                                                                                               |  | 6                                                                                |  | Recommended Scoring<br>(10 Total Points) | Consider following student work to determine if a mistake is a minor miscalculation, a major error, or a misconception.                                                      |
|                                                                                                                                                        |  | $-13.026$                                                                        |  | Additional Information                   | Watch for student misconceptions based on place value. Students are not likely to determine the sum using fractions, but they should have the freedom to do so if they wish. |

| Question                                                                                                                                                                                                                                          | 7                                                                                                                                                                                                        | Benchmark(s)                                               | MA.7.NSO.2.2                                                                                                            |
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| Using Fractions                                                                                                                                                                                                                                   | Using Decimals                                                                                                                                                                                           | <b>Recommended Scoring</b><br><br><b>(10 Total Points)</b> | Consider following student work to determine if a mistake is a minor miscalculation, a major error, or a misconception. |
| $-3\frac{3}{20} \div \left(\frac{7}{10}\right)$ $-\frac{63}{20} \div \frac{7}{10}$ $-\frac{63}{20} \cdot \frac{10}{7}$  $-\frac{9}{2} \text{ or any equivalent}$ | $-3\frac{3}{20} \div \left(\frac{7}{10}\right)$ $-3.15 \div 0.7$ $0.7 \overline{)3.15}$ $07.\overline{)31.5}$  $-4.5$ |                                                            |                                                                                                                         |
|                                                                                                                                                                                                                                                   |                                                                                                                                                                                                          | <b>Additional Information</b>                              | The key shows students using common factors to "simplify" the answer. Students are not required to do this.             |

| Question                                                                                                                                         | 8                                                                                                                                                                                                                                          | Benchmark(s)                                               | MA.7.NSO.2.2                                                                                                                                                                                                              |
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| Fractions                                                                                                                                        | Decimals                                                                                                                                                                                                                                   | <b>Recommended Scoring</b><br><br><b>(10 Total Points)</b> | Consider following student work to determine if a mistake is a minor miscalculation, a major error, or a misconception.                                                                                                   |
| $7\frac{3}{4} \cdot \left(-\frac{15}{8}\right)$ $\frac{31}{4} \cdot \frac{-15}{8}$ $-\frac{465}{32} = -14\frac{17}{32}$ <p>or any equivalent</p> | $7\frac{3}{4} \cdot \left(-\frac{15}{8}\right)$ $7\frac{3}{4} \cdot \left(-1\frac{7}{8}\right)$ $7.75 \cdot -1.875$ $\begin{array}{r} 1.875 \\ \times 7.75 \\ \hline 9375 \\ 131250 \\ \hline 1312500 \\ 14.53125 \end{array}$ $-14.53125$ |                                                            |                                                                                                                                                                                                                           |
|                                                                                                                                                  |                                                                                                                                                                                                                                            | <b>Additional Information</b>                              | This question is an excellent example where students with flexible thinking may find using decimals are more efficient than fractions. Using a similar question for student discussion may help to build student fluency. |

| Question                                                                                                                                                                                   |  | 9                                                                                                                                                                             | Benchmark(s)                             | MA.7.NSO.2.2                                                                                                                                            |
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| Using Fractions                                                                                                                                                                            |  | Using Decimals                                                                                                                                                                | Recommended Scoring<br>(10 Total Points) | Consider following student work to determine if a mistake is a minor miscalculation, a major error, or a misconception.                                 |
| $3.14 \cdot 1\frac{2}{5}$<br><br>$3\frac{14}{100} \cdot \frac{7}{5}$<br><br>$\frac{314}{100} \cdot \frac{7}{5}$<br><br>$\frac{2198}{500} = 4\frac{99}{250}$<br><br>or any other equivalent |  | $3.14 \cdot 1\frac{2}{5}$<br><br>$\frac{2}{5} = 0.4$<br><br>$3.14 \cdot 1.4$<br><br>$\begin{array}{r} 3.14 \\ \times 1.4 \\ \hline 1256 \\ +3140 \\ \hline 4.396 \end{array}$ | Additional Information                   | This question is assessing student flexibility with different forms of rational numbers. Teachers should look for student understanding of equivalency. |

| Question | 10 | Benchmark(s)                                 | MA.7.NSO.2.2                                                                                                                                            |
|----------|----|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |    | Recommended Scoring<br><br>(10 Total Points) | Consider following student work to determine if a mistake is a minor miscalculation, a major error, or a misconception.                                 |
|          |    |                                              |                                                                                                                                                         |
|          |    | Additional Information                       | This question is assessing student flexibility with different forms of rational numbers. Teachers should look for student understanding of equivalency. |
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# Form C

| Question                                                                                                                                                                                                                                                                                                                                                                     | 1                                                                                                                                                                                                                                                                                                                                                                                 | Benchmark(s)                                                                                                                                        | MA.7.NSO.2.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div data-bbox="73 308 714 357">Using Long Division</div> $-52.3452 \div (-3.62)$ $\begin{array}{r} 3.62 \overline{) 52.3452} \\ 362 \overline{) 523452} \\ \underline{362} \phantom{00} \\ 1614 \phantom{00} \\ \underline{1448} \phantom{00} \\ 1665 \phantom{00} \\ \underline{1448} \phantom{00} \\ 2172 \phantom{00} \\ \underline{2172} \\ 0 \end{array}$ <p>14.46</p> | <div data-bbox="724 308 1354 357">Using Fractions</div> $-52.3452 \div (-3.62)$ $-52 \frac{3452}{10000} \div -3 \frac{62}{100}$ $\frac{-523452}{10000} \div \frac{-362}{100}$ $\frac{-523452}{10000} \cdot \frac{100}{-362}$ $\frac{-52345200}{-3620000}$ $\frac{52345200}{3620000} \text{ or } \frac{523452}{36200} \text{ or } \frac{1446}{100}$ <p>or any other equivalent</p> | <div data-bbox="1396 292 1669 487">Recommended Scoring<br/>(10 Total Points)</div> <div data-bbox="1417 974 1648 1055">Additional Information</div> | <p>Consider following student work to determine if a mistake is a minor miscalculation, a major error, or a misconception.</p> <p>The two methods shown are representative of how flexible the benchmark allows students to be. Teachers often ask about the expectations of statewide assessments. The draft test item specifications do not require that answers be in the most simplified form so students should be able to answer equation editor questions without simplification. Multiple choice questions may require students to recognize equivalent responses which means students are applying their knowledge from MA.7.NSO.1.2. Since this is requiring knowledge of two different benchmarks it is not likely students would have to simplify when a question is assessing their knowledge of MA.7.NSO.2.2 because Florida's assessments assess one benchmark for each question.</p> |

| Question                                                                                                                                                                                                   |  | 2                                                                                                                                                                                                             | Benchmark(s)                                                | MA.7.NSO.2.2                                                                                                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>Using Fractions</div> $-2.7 \cdot \frac{16}{5}$ $-2 \frac{7}{10} \cdot \frac{16}{5}$ $\frac{-27}{10} \cdot \frac{16}{5}$ $\frac{-432}{50} \text{ or } -\frac{216}{25}$ <p>or any other equivalent</p> |  | <div>Using Decimals</div> $-2.7 \cdot \frac{16}{5}$ $\frac{16}{5} = 3 \frac{1}{5}$ $\frac{1}{5} = 0.20$ $-2.7 \cdot 3.2$ $\begin{array}{r} 2.7 \\ \times 3.2 \\ \hline 54 \\ 810 \\ \hline -8.64 \end{array}$ | <div>Recommended Scoring</div> <div>(10 Total Points)</div> | Consider following student work to determine if a mistake is a minor miscalculation, a major error, or a misconception.                                 |
|                                                                                                                                                                                                            |  |                                                                                                                                                                                                               | <div>Additional Information</div>                           | This question is assessing student flexibility with different forms of rational numbers. Teachers should look for student understanding of equivalency. |

| Question                                                                                                                                                                                                                                              | 3                                                                                                                                                                                           | Benchmark(s)                                                                        | MA.7.NSO.2.2                                                                                                                                                                        |
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| <div data-bbox="228 167 558 207">Using Fractions</div> $8\frac{2}{9} + \left(-\frac{25}{6}\right)$ $\frac{74}{9} - \frac{25}{6}$ $\frac{148}{18} - \frac{75}{18}$ <div data-bbox="180 659 611 732"><math>\frac{73}{18}</math> or any equivalent</div> | <div data-bbox="873 167 1199 207">Using Decimals</div> $8\frac{2}{9} + \left(-\frac{25}{6}\right)$ $8.\bar{2} - 4.1\bar{6}$ <div data-bbox="995 513 1077 553">4.0<math>\bar{5}</math></div> | <div data-bbox="1400 204 1667 394">Recommended Scoring<br/>(10 Total Points)</div>  | Consider following student work to determine if a mistake is a minor miscalculation, a major error, or a misconception.                                                             |
|                                                                                                                                                                                                                                                       |                                                                                                                                                                                             | <div data-bbox="1423 573 1640 646">Additional Information</div>                     | Two possible methods are shown.                                                                                                                                                     |
| Question                                                                                                                                                                                                                                              | 4                                                                                                                                                                                           | Benchmark(s)                                                                        | MA.7.NSO.2.2                                                                                                                                                                        |
| <div data-bbox="636 1141 793 1182">-22.183</div>                                                                                                                                                                                                      |                                                                                                                                                                                             | <div data-bbox="1400 889 1667 1079">Recommended Scoring<br/>(10 Total Points)</div> | Consider following student work to determine if a mistake is a minor miscalculation, a major error, or a misconception.                                                             |
|                                                                                                                                                                                                                                                       |                                                                                                                                                                                             | <div data-bbox="1423 1263 1640 1336">Additional Information</div>                   | Watch for student misconceptions based on place value. Students are not likely to determine the difference using fractions, but they should have the freedom to do so if they wish. |



| Question                                                                                                                                                     | 5                                                                                                                                                                                                                                      | Benchmark(s)                                                                                                                                                                      | MA.7.NSO.2.2                                                                                                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div data-bbox="289 191 491 232">Fractions</div> $-12\frac{5}{8} \cdot 2\frac{1}{4}$ $-\frac{101}{8} \cdot \frac{9}{4}$ $-\frac{909}{32} = -28\frac{13}{32}$ | <div data-bbox="940 191 1134 232">Decimals</div> $-12\frac{5}{8} \cdot 2\frac{1}{4}$ $-12.625 \cdot 2.25$ $\begin{array}{r} 12.625 \\ \times 2.25 \\ \hline 63125 \\ 252500 \\ \hline 2525000 \\ 28.40625 \\ \\ -28.40625 \end{array}$ | <div data-bbox="1400 203 1663 394"> <b>Recommended Scoring</b><br/><br/> <b>(10 Total Points)</b> </div> <div data-bbox="1421 657 1642 730"> <b>Additional Information</b> </div> | <div data-bbox="1705 167 2051 433"> Consider following student work to determine if a mistake is a minor miscalculation, a major error, or a misconception. </div> <div data-bbox="1705 657 2051 730"> Two possible methods are shown. </div> |

| Question                                                                                                                                                                                                                                                                                                 | 6                                                                                                                                                       | Benchmark(s)                                                                                                                                            | MA.7.NSO.2.2                                                                                                                                                                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div data-bbox="71 196 709 245">Using Fractions</div> $6.17 + \left(-2\frac{3}{4}\right)$ $2\frac{3}{4} = 2\frac{75}{100}$ $6.17 = 6\frac{17}{100}$ $6\frac{17}{100} - 2\frac{75}{100}$ $\frac{617}{100} - \frac{275}{100}$ $\frac{342}{100} \text{ or } 3\frac{42}{100}$ <p>or any other equivalent</p> | <div data-bbox="724 196 1356 245">Using Decimals</div> $6.17 + \left(-2\frac{3}{4}\right)$ $2\frac{3}{4} = 2\frac{75}{100} = 2.75$ $6.17 - 2.75$ $3.42$ | <div data-bbox="1375 196 1688 435"> Recommended Scoring<br/>(10 Total Points) </div> <div data-bbox="1375 435 1688 1206"> Additional Information </div> | <p>Consider following student work to determine if a mistake is a minor miscalculation, a major error, or a misconception.</p> <p>This question is assessing student flexibility with different forms of rational numbers. Teachers should look for student understanding of equivalency.</p> |

| Question                                                                                                                                                                                                        | 7                                                                                                                                                                                       | Benchmark(s)                                                                        | MA.7.NSO.2.2                                                                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div data-bbox="226 175 558 215">Using Fractions</div> $-8\frac{2}{9} - \left(\frac{13}{6}\right)$ $-\frac{74}{9} - \frac{13}{6}$ $-\frac{148}{18} - \frac{39}{18}$ $-\frac{187}{18} \text{ or any equivalent}$ | <div data-bbox="873 175 1199 215">Using Decimals</div> $-8\frac{2}{9} - \left(\frac{13}{6}\right)$ $-8\frac{2}{9} - \left(2\frac{1}{6}\right)$ $-8.\bar{2} - 2.1\bar{6}$ $-10.3\bar{8}$ | <div data-bbox="1398 204 1665 394">Recommended Scoring<br/>(10 Total Points)</div>  | Consider following student work to determine if a mistake is a minor miscalculation, a major error, or a misconception.                                                             |
|                                                                                                                                                                                                                 |                                                                                                                                                                                         | <div data-bbox="1421 574 1642 646">Additional Information</div>                     | Two possible methods are shown.                                                                                                                                                     |
| Question                                                                                                                                                                                                        | 8                                                                                                                                                                                       | Benchmark(s)                                                                        | MA.7.NSO.2.2                                                                                                                                                                        |
| $-43.133$                                                                                                                                                                                                       |                                                                                                                                                                                         | <div data-bbox="1398 899 1665 1089">Recommended Scoring<br/>(10 Total Points)</div> | Consider following student work to determine if a mistake is a minor miscalculation, a major error, or a misconception.                                                             |
|                                                                                                                                                                                                                 |                                                                                                                                                                                         | <div data-bbox="1421 1268 1642 1339">Additional Information</div>                   | Watch for student misconceptions based on place value. Students are not likely to determine the difference using fractions, but they should have the freedom to do so if they wish. |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 9                                                                                                                                                                                                                                                                                                                                                   | Benchmark(s)                                                                                                                                                                      | MA.7.NSO.2.2                                                                                                                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div data-bbox="228 183 558 224">Using Fractions</div> $-2\frac{9}{20} \div \frac{7}{4}$ $-\frac{49}{20} \div \frac{7}{4}$ $-\frac{49}{20} \cdot \frac{4}{7}$ <div data-bbox="268 690 493 868"> <math display="block"> \begin{array}{r} \overset{7}{\cancel{49}} \overset{1}{\cancel{4}} \\ -\frac{\cancel{20}}{\cancel{7}} \\ \hline \underset{5}{-1} \underset{1}{\frac{2}{5}} \end{array} </math> </div> <p><math>-\frac{7}{5} = -1\frac{2}{5}</math> or any equivalent</p> | <div data-bbox="873 183 1199 224">Using Decimals</div> $-2\frac{9}{20} \div \frac{7}{4}$ $-2\frac{9}{20} \div 1\frac{3}{4}$ $-2.45 \div 1.75$ $1.75 \overline{)2.45}$ $175 \overline{)245.}$ $ \begin{array}{r} 1.4 \\ 175 \overline{)245.0} \\ \underline{-175} \phantom{0} \\ 700 \\ \underline{-700} \\ 0 \end{array} $ <p><math>-1.4</math></p> | <div data-bbox="1398 204 1665 394"> <b>Recommended Scoring</b><br/><br/> <b>(10 Total Points)</b> </div> <div data-bbox="1419 883 1642 954"> <b>Additional Information</b> </div> | <p>Consider following student work to determine if a mistake is a minor miscalculation, a major error, or a misconception.</p> <p>The key shows students using common factors to "simplify" the answer. Students are not required to do this.</p> |

| Question                                                                                                                                | 10                                                                                                                                                                                            | Benchmark(s)                                 | MA.7.NSO.2.2                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                         |                                                                                                                                                                                               | Recommended Scoring<br><br>(10 Total Points) | Consider following student work to determine if a mistake is a minor miscalculation, a major error, or a misconception. All equivalent answers should be expected. The possible methods shown are likely the most common methods that students use. But students who choose other methods and correctly determine an equivalent value should not lose points. |
| Using Decimals                                                                                                                          | Using Fractions                                                                                                                                                                               |                                              |                                                                                                                                                                                                                                                                                                                                                               |
| $-3.8 \cdot 0.571$<br><br>$\begin{array}{r} 0.571 \\ \times 3.8 \\ \hline 4568 \\ +17130 \\ \hline 2.1698 \end{array}$<br><br>$-2.1698$ | $-3.8 \cdot 0.571$<br><br>$-\frac{38}{10} \cdot \frac{571}{1000}$<br><br>$\frac{-38}{10} \cdot \frac{571}{1000} = \frac{-21698}{10000}$<br><br>$\frac{-10849}{5000}$<br><br>or any equivalent | Two possible methods are shown.              |                                                                                                                                                                                                                                                                                                                                                               |